

ADHAM EHAB ERFAN, B.Sc.

Communication & Electronics Engineer | Wireless, RF, Embedded Systems & Automation
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PROFILE

Communication and Electronics Engineering graduate specializing in the intersection of wireless networks and deep learning. With hands-on research experience in MIMO systems and neural codebook design, I am seeking a Master's/PhD scholarship to drive advanced, research-intensive projects in wireless communication, 5G/6G signal processing, deep learning and communication theory, or any other related fields.

CORE EXPERTISE

- MIMO & Antenna Systems • 5G NR Physical Layer (SSB/CSI-RS) • Channel Estimation & MMSE Equalization
- Beamforming & Codebook Design • Spectrum & Propagation Analysis • Multi-path Channel Modelling
- Interference Management • Ray-Tracing Simulation (Sionna) • MATLAB RF / Comms Toolboxes
- Embedded RF (ESP32, ARM) • Technical Documentation

EDUCATION

B.Sc. in Communications & Electronics Engineering | Alexandria University | Egypt 2021 - 2026

GPA: 3.57/4.00

Graduation project (Excellent): Neural Codebook Design for MIMO Beam Management by building an end-to-end deep learning pipeline for SSB and CSI-RS codebook generation in multi-cell 5G networks, simulated using the Sionna ray-tracing library.

Thanawya Ama | Omar Ibn El Hattab | Egypt 2020 - 2021

Grade: Excellent **3rd in school**

Core Focus: Advanced Mathematics, Mechanics, and Physics

EXPERIENCE

English Instructor 2025 – Present

IEEE SSCS | Hybrid

- Designed and delivered an intensive 8-week Academic IELTS preparation curriculum, specifically structured to help students achieve a target band score of 7.0 or higher.
- Instructed comprehensive, strategy-driven modules across all four core competencies: reading, listening, writing, and speaking.
- Equipped students with advanced test-taking techniques, including time-management strategies for complex reading passages, structural frameworks for essay writing, and fluency tactics for speaking interviews.
- Formulated a continuous evaluation system using targeted quizzes and full-length mock exams to benchmark student progress, identify individual weaknesses, and provide actionable, personalized feedback.

AI Trainer 2025 – Present

RWS | Remote

- Cleaning datasets, doing transcription for audios, and translating EN/AR .

AI Model Developer 2024 – 2025

Outlier | Remote

- Evaluated AI outputs in mathematics and algorithm design; provided structured RLHF feedback.

Mathematics Teacher 2023 – 2024

Center | On-site

- Taught high school students (first and second level).

TRAINING & INTERNSHIPS

R&D 2024 – 2026

University Center for Career Development | Alexandria, Egypt

- Contributed to technical documentation, research, and data analysis.

Embedded Systems Intern

Summer 2024

Siemens | Cairo, Egypt

- Developed startup code and peripheral drivers for AVR and ARM Cortex-M microcontrollers; performed memory mapping and low-level hardware bring-up.
- Implemented and debugged UART, SPI, I2C serial protocols for sensor interfacing and inter-device communication in an industrial environment.
- Programmed ESP32 for real-time wireless data acquisition and cloud streaming; practiced hardware–software integration with oscilloscope and logic analyzer.

Networks Training (CCNA)

Summer 2023

ICG | Alexandria, Egypt

- Designed and simulated a multi-subnet LAN in Cisco Packet Tracer utilizing RIP/OSPF protocols to enable dynamic routing between distinct gateways.

TECHNICAL SKILLS

Expert

Advanced

Intermediate

Tools

MATLAB, DSP Toolbox, RF Toolbox, Simulink, MIMO Systems, Beamforming
Python, Signal Processing, SDR, Machine Learning (NumPy, TensorFlow), CNNs, Autoencoders, DQN
C, C++, Embedded Systems (AVR, ARM), UART/SPI, IoT Development
LATEX, Microsoft office, Proteus, Multisim, OrCAD, ROS, Git, AutoCAD

LANGUAGES

English Advanced B2

French Basic A1

CERTIFICATES

- **IELTS Academic** — Band 6.0
- **Deep Learning** — NVIDIA
- **Machine Learning** — Generative Technology Center
- **Embedded Systems Essentials** — SIEMENS
- **Embedded Systems**— ITI

EVENTS

- **Job Fair**— Faculty of Engineering, Alexandria University
- **Technical Summit** — Bibliotheca Alexandrina
- **ICT Exhibition** — Cairo (International Communications & Technology)
- **Cairo International Book Fair**— Cairo

SOFT SKILLS

Drawing, Time management, Teamwork, Creativity, Critical thinking, Adaptability.

PERSONAL PROJECT & APPLICATIONS

Islamic MVP — Mobile Application— Prayer Tracker

- Built a prayer time tracker application with daily scheduling, notifications, and progress logging.

MailWatch — Mobile Application— Gmail & WhatsApp Notifier

- Built an app that scans the user's inbox and spam folder for specified emails from the user and notifies on WhatsApp.

Pharmacy Data Entry App — Mobile Application— Healthcare Data Management

- Designed and built a data entry and record management system for pharmacy inventory and patient prescription logging.